



EUROPEAN COMMISSION 2026

Covariates in DiD

OLS specs, IPW, OR, DR

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The standard TWFE with additive X

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Worry 1: β changes at the policy break

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post.

Worry 2: treatment effect varies with X

$$Y_{it} = \alpha_i + \lambda_t + \delta_0 (D_i \times \text{Post}_t) + \tau (D_i \times \text{Post}_t \times X_i) + \beta X_i + \varepsilon_{it}$$

$$\text{ATT} = \frac{1}{n_T} \sum_{D_i=1} (\hat{\delta}_0 + \hat{\tau} X_i)$$

Both worries: the fully saturated spec

$$Y_{it} = \alpha_i + \lambda_t + \delta_0 (D_i \times \text{Post}_t) + \boldsymbol{\tau} (D_i \times \text{Post}_t \times X_i) + \boldsymbol{\gamma} (\text{Post}_t \times X_i) + \boldsymbol{\beta} X_i + \varepsilon_{it}$$

$$\text{ATT} = \frac{1}{n_T} \sum_{D_i=1} (\hat{\delta}_0 + \hat{\boldsymbol{\tau}} X_i) \quad \cdot \quad \text{This is **Spec C** in the lab.}$$

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

$$\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i) \quad (\text{propensity score}) \quad \Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i .
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i)) / (1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{ATT} = \sum w_i \Delta Y_i / n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \rightarrow 1$ for a control unit, its weight $\rightarrow \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR: model the trend, impute the counterfactual

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{D_i=1} \left[\Delta Y_i - \hat{\mu}_0(X_i) \right]$$

$$\hat{\mu}_0(X_i) = \hat{E}[\Delta Y \mid D = 0, X] \quad \text{estimated on controls only}$$

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** $\text{ATT} = \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

DR: two chances to be right

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
✓	✓	✓
✓	×	✓
×	✓	✓
×	×	No

“Doubly robust” $\neq$ doubly unbeatable.

Lab: ten estimators, one benchmark

Labs/Lalonde-Covariates/lalonde_all_specs.do

NSW treated + CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **\$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B ($\text{Post} \times X$) · Spec C (Saturated)

OR by hand · OR drdid · IPW by hand · IPW drdid · DR by hand · DR drdid

A Monte Carlo to rank the estimators

Estimators: TWFE, OR, IPW, DR

DGPs: 4 (vary whether OR model and pscore are correct)

n : 1,000 Replications: 10,000

Report: Bias, RMSE, SE, coverage, CI length

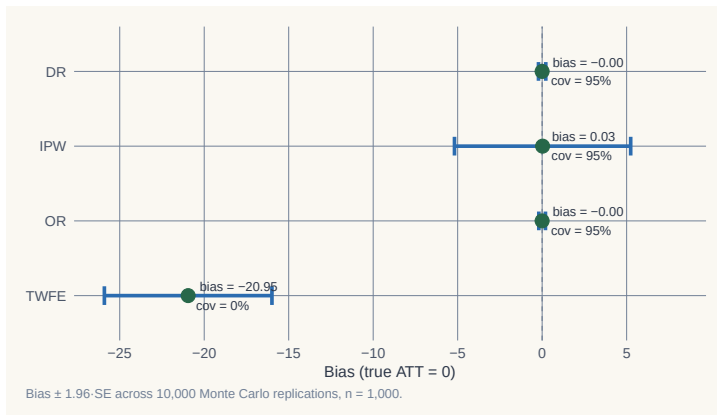
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	−20.95	21.12	2.53	0.000	9.91
OR	−0.001	0.10	0.10	0.950	0.40
IPW	+0.026	2.77	2.66	0.952	10.44
DR	−0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

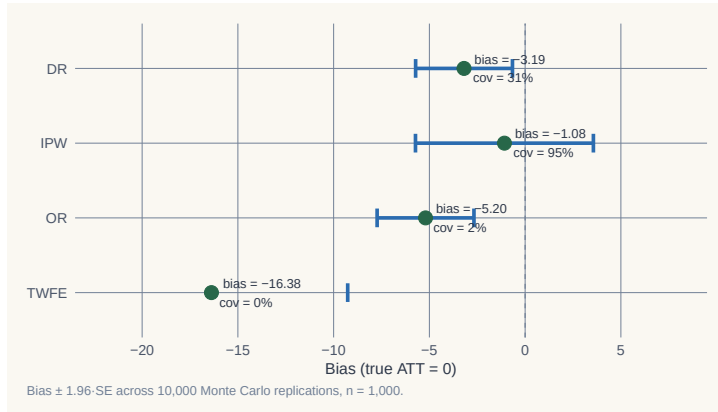


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	−16.38	16.54	3.63	0.000	14.22
OR	−5.20	5.36	1.29	0.015	5.05
IPW	−1.08	2.66	2.37	0.949	9.31
DR	−3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X** : coverage zero in every DGP. **Stop using it as a default.**
- **OR**: efficient when right. Coverage 1.5% when wrong.
- **IPW**: less efficient, but robust to outcome misspecification.
- **DR**: best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—
from **“does PT hold?”**
to **“is my model of $Y(0)$ or $p(X)$ correct?”**

You still have to answer it.

The question is never

“did I control for X ?”

The question is always

“did I get the model right?”
